[PHYS 6268] Nonlinear Dynamics: Problem Set 6

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Strogatz Book: Problems 6.5.6, 6.6.3, 6.6.7, & 6.7.1

1 Problem 6.5.6: Epidemic model revisited

In Exercise 3.7.6, you analyzed the Kermack-McKendrick model of an epidemic by reducing it to a certain first-order system. In this problem, you'll see how much easier the analysis becomes in the phase plane. As before, let $x(t) \geq 0$ denote the size of the healthy population and $y(t) \geq 0$ denote the size of the sick population. Then the model is,

$$\dot{x} = -kxy \quad , \quad \dot{y} = kxy - ly \qquad ,$$

where $k, l > 0$. The equation for $z(t)$, the number of deaths, plays no role in the dynamics in the x, y plane so we omit it.

1.1 (a) Find and classify all the fixed points.

We begin with the system

$$\begin{aligned} \dot{x} &= -kxy \\ \dot{y} &= kxy - ly \qquad , \end{aligned}$$

subject to the constraint that the total population does not change (including those that have deceased). This assumption corresponds to the fact that the outbreak of the transmitted pathogen is rapid on the timescale of population dynamics (such as changing birth/death rates). Hence, this condition (introducing $z(t)$ to represent the total of the deceased populus) becomes:

$$\begin{aligned} N(t) &= x(t) + y(t) + z(t) \\ \dot{N}(t) &= 0 \\ \implies \dot{x} + \dot{y} + \dot{z} &= 0 \\ \implies \dot{z} &= -ly \quad . \end{aligned}$$

To determine the fixed points, we search for solutions (x^*, y^*) that correspond to $\dot{x} = \dot{y} = 0$. This clearly occurs at the origin $(x^*, y^*) = (0, 0)$. However, due to the fact that each term in $\dot{x}$ or $\dot{y}$ is proportional to y, the line $y = 0$ represents an entire *set* of fixed points. Hence, the fixed points explicitly are:

$$(x^*, y^*) \longrightarrow \quad (s, 0) \quad (s \in \mathbb{R}) \quad .$$

with special cases at $s = 0$ and $s = l/k$. To assess the stability of these fixed points, we linearize the system:

$$\frac{\mathrm{d}}{\mathrm{d}t}\begin{pmatrix} x \\ y \end{pmatrix} = \dot{\mathbf{x}} = \mathbb{A} \cdot \mathbf{x} \quad ; \quad \mathbb{A} \equiv \begin{pmatrix} \partial_x \dot{x} & \partial_y \dot{x} \\ \partial_x \dot{y} & \partial_y \dot{y} \end{pmatrix} \quad ,$$

where $\mathbb{A}$ can be directly calculated from the system. Differentiation yields

$$\mathbb{A} = \begin{pmatrix} -ky & -kx \\ ky & kx - l \end{pmatrix} \quad ,$$

which, evaluated at the fixed point $(x^*, y^*) = (s, 0)$, becomes

$$\mathbb{A} = \begin{pmatrix} 0 & -ks \\ 0 & ks - l \end{pmatrix} \quad .$$

We first consider the case that $s = 0$. This yields

$$\mathbb{A} = \begin{pmatrix} 0 & 0 \\ 0 & -l \end{pmatrix} \quad ,$$

which implies that $\det \mathbb{A} = 0$ and $\operatorname{Tr} \mathbb{A} = -l$. Similar to the treatment in the previous problem set, the eigenvalues of the system can be written in terms of the trace ($\operatorname{Tr} \mathbb{A}$) and determinant ($\det \mathbb{A}$) of $\mathbb{A}$, such that

$$\lambda_{\pm} = \frac{\operatorname{Tr} \mathbb{A} \pm \sqrt{\operatorname{Tr} \mathbb{A}^2 - 4 \det \mathbb{A}}}{2} \quad , \tag{1}$$

allowing us to classify the system using exclusively $\operatorname{Tr} \mathbb{A}$ and $\det \mathbb{A}$. For this case, $\det \mathbb{A} = 0 \implies \lambda_2 = 0$ immediately. Via $\operatorname{Tr} \mathbb{A} = \lambda_1 + \lambda_2$, this implies immediately that $\lambda_1 = -l$. The eigenvectors are simply the canonical basis vectors of the Cartesian system: $\mathbf{v}_1 = \hat{\mathbf{y}}$ and $\mathbf{v}_2 = \hat{\mathbf{x}}$. The fact that $\det \mathbb{A} = 0$ implies that we have a non-isolated fixed point. As determined via the above analysis, there is an entire line of fixed points. Since $\operatorname{Tr} \mathbb{A} < 0$ for $s = 0$, this line is stable (attracting) from the y-direction at the origin. The x coordinate is always fixed, since $\dot{x} \propto y = 0$.

Now we consider the more general case for arbitrary s in order to classify the entire line of fixed points. In this way, $\operatorname{Tr} \mathbb{A} = ks - l$ and $\det \mathbb{A} = 0$ (still). Therefore, the line of fixed points along $y = 0$ are attracting for all $s = x < l/k$, since the trace of the stability matrix is less than zero. However, at this critical point, the system transforms and the fixed point at the origin has a reversed stability along the y-direction. For $x > l/k$, the fixed points are unstable, and the $y = 0$ line becomes a source of trajectories.

1.2 (b) Sketch the nullclines and the vector field.

The nullclines are given by:

$$-kxy = 0$$

and

$$kxy - ly = 0 \quad .$$

Hence, the resultant nullcline lines are $x = 0$, $y = 0$, and $x = l/k$.

These nullclines along with the vector field corresponding to the system given in section 1 are shown in Figure 1. In this figure, the vector field has arrows with a length proportional to the local magnitude of the gradient in the $x - y$ plane.

1.3 (c) Find a conserved quantity for the system.

In order to find a conserved quantity, the typical strategy is to rely on some physical context of the system to write down an expression for the total system energy (i.e., a Hamiltonian). This expression is invariant along trajectories since energy is conserved in closed, dissipation-free systems, and its change is equal to the work done by the dissipation should it be present. However, it is conceptually tricky to formulate an expression for the "energy" associated with an epidemic system, so we take a less inspired and more brute-force approach.

A conserved quantity $E(x, y)$ is a function of system variables that is constant along any given trajectory in phase space for a dynamical system. This can be formulated mathematically as

$$\frac{\mathrm{d}}{\mathrm{d}t} E(x, y) = 0 \quad ,$$

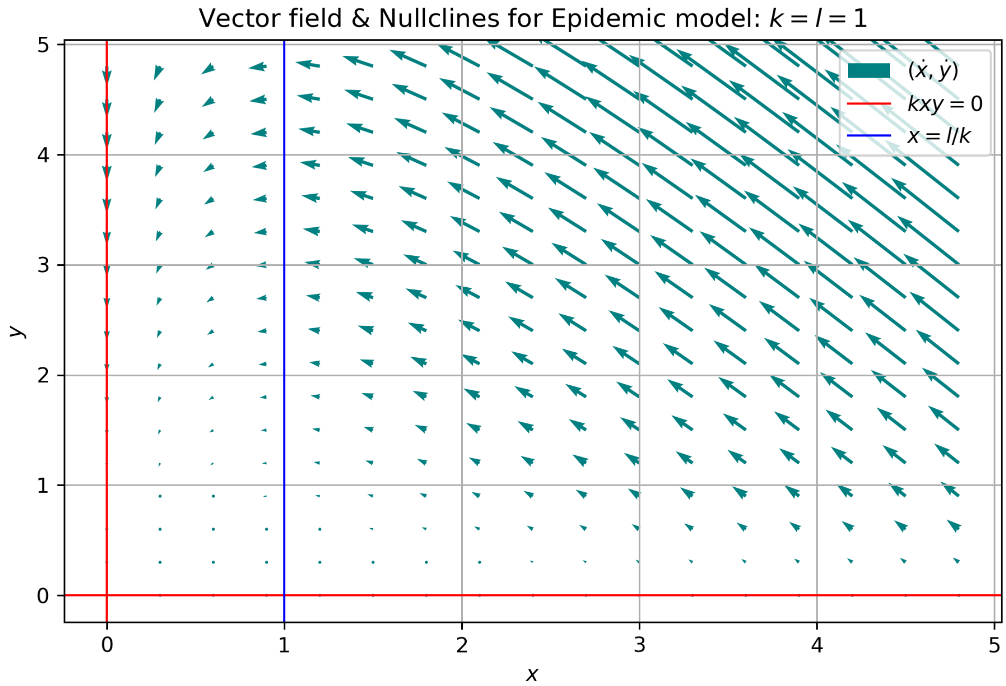

Figure 1: Vector field (teal) and nullclines (red, blue) depicted for the system described by the epidemic model in section 1. The length of the vectors are linearly proportional to their magnitude.

where $\frac{\mathrm{d}}{\mathrm{d}t}$ is the *total* derivative of E. Thus, this really states that

$$\frac{\partial E}{\partial x}\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{\partial E}{\partial y}\frac{\mathrm{d}y}{\mathrm{d}t} = 0 \qquad .$$

This is just

$$\partial_x E\, \dot{x} + \partial_y E\, \dot{y} = 0 \qquad ,$$

which provides us a partial differential equation for our conserved quantity $E(x, y)$. First, we substitute the expressions for $\dot{x}, \dot{y}$ from our system:

$$(-kxy)\, \partial_x E + (kxy - ly)\, \partial_y E = 0 \qquad .$$

Then, we can rearrange

$$\implies \partial_y E = \partial_x E \left(\frac{kxy}{kxy - ly} \right)$$
$$\implies \partial_y E = \partial_x E \left(\frac{1}{1 - \frac{ly}{kxy}} \right)$$
$$\implies \partial_y E = \partial_x E \left(\frac{1}{1 - l/kx} \right).$$

It is helpful to define

$$\frac{\partial E}{\partial y} = \frac{\partial E}{\partial x} \left(\frac{1}{1 - l/kx} \right) = \frac{\partial E}{\partial x}\, h(x) \qquad ,$$

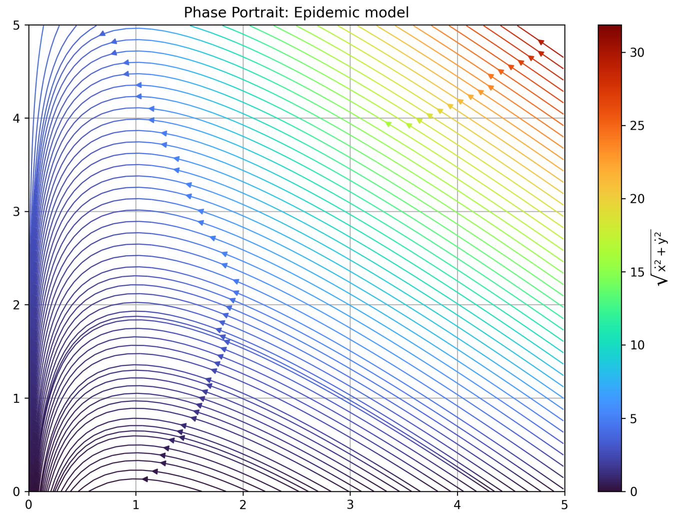

Figure 2: Phase portrait for Epidemic model of section 1. The parameters used for the calculation of this phase portrait are $k = l = 1$. The y coordinate is represented by the vertical axis, and x by the horizontal axis.

where $h(x) \equiv \left(\frac{1}{1-l/kx}\right)$.

Now, we are still faced with a PDE. We first make the assumption that the expression for E can be written as

$$E(x, y) = X(x) + Y(y) \qquad ,$$

to simplify the problem. This then suggests that

$$\frac{\partial E}{\partial y} = \partial_y Y = \frac{\partial E}{\partial x} h(x) = \partial_x X h(x) \qquad ,$$

where we take advantage of the (assumed) separability of E. Now, we make one final assumption and fix $Y(y) = y$. This implies that

$$\partial_x X\, h(x) = 1 \qquad \Longrightarrow \qquad \partial_x X = \frac{1}{h(x)} \qquad .$$

Hence, this gives us an expression that can be directly integrated for E:

$$\Longrightarrow \; E(x, y) = y + \int h^{-1}(x)\, \mathrm{d}x = y + \int \left(1 - \frac{l}{kx}\right) \mathrm{d}x \qquad .$$

Doing so yields

$$E(x, y) = y + x - \frac{l}{k} \ln |x| \qquad .$$

Hence, the quantity above is a conserved quantity: a real-valued, continuous function $E(x, y)$ that is constant along trajectories of the system.

1.4 (d) Plot the phase portrait. What happens as $t \longrightarrow \infty$?

The phase portrait for the case $k = l = 1$ is given in Figure 2. The magnitude of the gradient is represented by the colormap.

As $t \longrightarrow \infty$, the system tends towards one of two fates. The dynamics of the system are determined by the ratio of the sick rate to the death rate, scaled by the initial population size. If the death rate l is high compared to the sick rate k, then the system will tend towards $y = 0$ and $x \approx N$ from the start, without reaching a maximum in disease spread (i.e., the system is "improving" from the beginning). However, if the sick rate (scaled by the initial healthy population) is high compared to the death rate, then the value of $y(t)$ grows rapidly *before* ever turning around and returning to zero. Thus, in either case, the system will tend toward $y(t) = 0$, and the values of the constants determine whether $x(t \longrightarrow \infty) \approx N$ or whether z tends towards a non-negligible value, leaving $x(t \longrightarrow \infty) = N - z(t \longrightarrow \infty)$.

1.5 (e) Let (x_0, y_0) be the initial condition. An *epidemic* is said to occur if $y(t)$ increases initially. Under what condition does an epidemic occur?

The condition for the epidemic can be written as $\dot{y}(t = 0) > 0$. In terms of $x_0, y_0 > 0$, we can write

$$\dot{x}(0) = -kx_0y_0 < 0 \qquad ,$$

and

$$\dot{y}(0) = kx_0y_0 - ly_0 \qquad .$$

Thus, it is evident that $\dot{y}(0) > 0$ if and only if

$$kx_0 > l \qquad ,$$

equivalent to the condition that we identified in the previous exercise where we used N to write the system in a single dimension. This is also equivalent to the threshold for different qualitative behaviors in the phase portrait discussed above.

2 Problem 6.6.3: Wallpaper

Consider the system

$$\dot{x} = \sin y \qquad , \qquad \dot{y} = \sin x \quad .$$

2.1 (a) Show that the system is reversible.

To show that the system is reversible, we demonstrate that it is invariant under the transformation $t \mapsto -t$ and $y \mapsto -y$. By demonstrating that the resulting system is equivalent, we show reversibility for the system described in section 2. That is,

$$\dot{x} = \frac{\mathrm{d}x}{\mathrm{d}t} \mapsto -\frac{\mathrm{d}x}{\mathrm{d}t} = -\dot{x} = \sin y \mapsto \sin(-y) = -\sin y$$

so we recover

$$\implies \dot{x} = \sin y \qquad .$$

Moreover,

$$\dot{y} = \frac{\mathrm{d}y}{\mathrm{d}t} \mapsto \frac{-\mathrm{d}y}{-\mathrm{d}t} = \frac{\mathrm{d}y}{\mathrm{d}t} = \sin x \mapsto \sin x \qquad ,$$

or

$$\dot{y} = \sin x \qquad .$$

Hence, we recover the same equations for $\dot{x}, \dot{y}$ as above under the transform $t \mapsto -t$ and $y \mapsto -y$, so the system is reversible.

2.2 (b) Find and classify all the fixed points.

Fixed points occur when

$$\dot{x} = \dot{y} = 0 \implies \sin x = \sin y = 0 \qquad ,$$

which immediately implies that

$$(x^*, y^*) = (m\pi, n\pi) \qquad m, n \in \mathbb{Z} \quad .$$

Therefore, since sin has a countably infinite set of zeroes, we must treat different general qualitative cases (e.g., m is even and n is odd) based on the different values of the stability matrix.

With the fixed points identified, we linearize the system about the fixed points using the matrix of stability. Let $\mathbf{x}$ be a column vector containing x, y. Then, linearization of the system is given by

$$\frac{\mathrm{d}}{\mathrm{d}t}\begin{pmatrix} x \\ y \end{pmatrix} = \dot{\mathbf{x}} = \mathbb{A}\cdot\mathbf{x} + \mathcal{O}(\mathbf{x}^2) \quad ; \quad \mathbb{A} \equiv \begin{pmatrix} \partial_x \dot{x} & \partial_y \dot{x} \\ \partial_x \dot{y} & \partial_y \dot{y} \end{pmatrix} \quad .$$

Calculating the partial derivatives yields

$$\mathbb{A} = \begin{pmatrix} 0 & \cos y \\ \cos x & 0 \end{pmatrix} \quad .$$

Therefore, at any given fixed point of the system, this matrix becomes

$$\mathbb{A} = \begin{pmatrix} 0 & \pm 1 \\ \pm 1 & 0 \end{pmatrix} \quad .$$

It can be seen immediately that for any fixed point $(x^*, y^*) = (m\pi, n\pi)$, the properties of the stability matrix $\mathbb{A}$ are given by

$$\mathrm{Tr}\,\mathbb{A} = 0 \quad ; \quad \det\mathbb{A} = -\cos(m\pi)\cos(n\pi) \quad .$$

From here, we can consider three separate cases.

(i) m is odd, n is odd
(ii) m is odd, n is even
(iii) m is even, n is even

We start with case (i): this implies that $\cos m\pi = \cos n\pi = -1$, which yields us $\det\mathbb{A} = -1$. Since $\det\mathbb{A} < 0$, the eigenvalues representing the linearized trajectories near x^*, y^* are real and opposite signs. Hence, the fixed point is a saddle, with the stable direction along lines $y = x + c$ for $c \in \mathbb{R}$. The eigenvalues are given by $\lambda = \pm 1$, with $\lambda_1 = +1$. The eigenvectors are then

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ -1 \end{pmatrix} , \qquad \mathbf{v}_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad .$$

Now, we move on to case (ii): this implies that $\mathrm{sgn}(\cos m\pi) \neq \mathrm{sgn}(\cos n\pi)$. Hence, $\det\mathbb{A} = 1 > 0$ and $\mathrm{Tr}\,\mathbb{A} = 0$ still, we now have new dynamics. The discriminant

$$\mathrm{Tr}\,\mathbb{A}^2 - 4\det\mathbb{A}$$

is less than zero with $\mathrm{Tr}\,\mathbb{A} = 0$, implying the fixed points are neutrally-stable centers. This can be verified by noting that the eigenvalues $\lambda = \pm i$ are complex conjugates.

Finally, we consider case (iii) where $\cos m\pi = \cos n\pi = 1$. Since $\det\mathbb{A} = -1$ and $\mathrm{Tr}\,\mathbb{A} = 0$, the eigendecomposition and dynamics of the system will be identical to that of case (i). Hence, for this case (analogous to case (i)), the fixed points are again saddles, with the stable direction along lines $y = x + c$ for $c \in \mathbb{R}$.

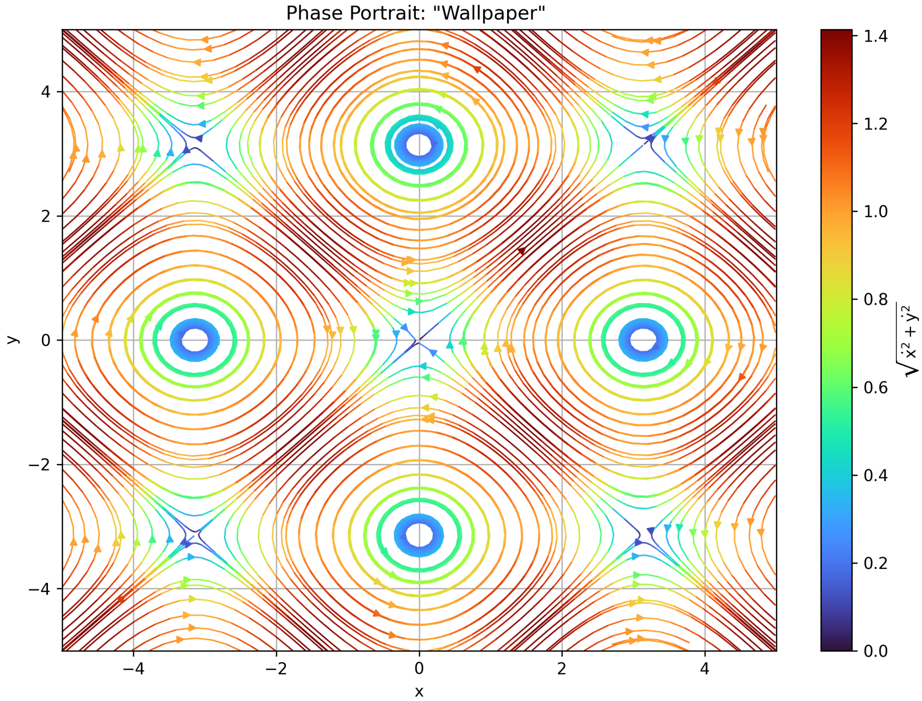

Figure 3: Phase portrait for the dynamical system given by $\dot{x} = \sin y$ and $\dot{y} = \sin x$. The horizontal axis shows the x coordinate and the vertical axis shows y. The color represents the magnitude of the gradient in the $x - y$ plane for the system.

2.3 (c) Show that the lines $y = \pm x$ are invariant (any trajectory that starts on them stays on them forever).

We have that $\dot{x} = \sin y$ and $\dot{y} = \sin x$. To show that the lines $y = \pm x$ are invariant, we first consider the positive case $y = x$:

$$\frac{\mathrm{d}x}{\mathrm{d}t} = \sin y = \sin x \quad ,$$

which is separable. This yields

$$\frac{\mathrm{d}x}{\sin x} = \mathrm{d}t \quad ,$$

which can be integrated directly:

$$\implies \log\left|\tan\frac{x}{2}\right| = t \quad .$$

Then, letting the constant of integration vanish (initial conditions are arbitrary), we have that

$$\implies x(t) = 2\arctan e^t \quad .$$

Similarly, $y(t) = 2\arctan e^t = x(t)$. In the alternative case that $y = -x$, we see that

$$\frac{\mathrm{d}x}{\mathrm{d}t} = \sin y = \sin -x = -\sin x \quad ,$$

which yields

$$\implies \log\left|\tan\frac{x}{2}\right| = -t \quad .$$

Then,

$$x(t) = 2\arctan e^{-t} \qquad .$$

2.4 (d) Sketch the phase portrait.

The phase portrait is depicted for the system in Figure 3. The periodic structure of the system is evident, with saddle points and centers with neutral stability visible in an alternating fashion, corresponding to the different cases of m, n discussed in section 2.2.

3 Problem 6.6.7: Oscillator with both positive and negative damping

Show that the system

$$\ddot{x} + x\dot{x} + x = 0$$

is reversible and plot the phase portrait.

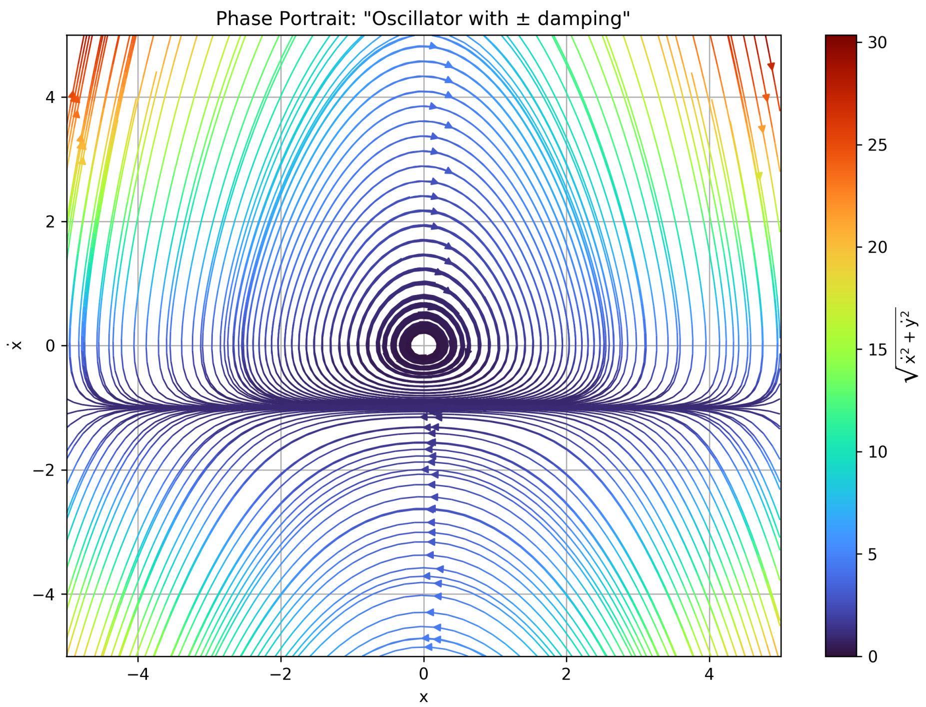

Figure 4: Phase portrait for the dynamical system given by $\ddot{x} + x\dot{x} + x = 0$. The horizontal axis shows the x coordinate and the vertical axis shows $\dot{x}$. The color indicates the magnitude of the gradient in the $(x - \dot{x})$ plane for the system.

To this end, we rewrite the second-order dynamical system as a coupled system of first-order equations. Let $\dot{x} = y$:

$$\dot{y} = \ddot{x} = -x(\dot{x}+1) = -x(y+1) = -xy - x$$
$$\dot{x} = y \qquad .$$

Following the method of reversibility used in section 2.1, we find that

$$\dot{x} \mapsto -\dot{x} = -y \mapsto y \qquad ,$$

which satisfies our criterion. However, considering the equation for $\dot{y}$, we find an inconsistency:

$$\dot{y} \mapsto \dot{y} \implies -xy - x \mapsto xy - x$$

which is a clear contradiction. Hence, we consider the reversibility under the transformations of each coordinate independently. First, we consider the full expression:

$$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + x\frac{\mathrm{d}x}{\mathrm{d}t} + x = 0 \qquad .$$

Now, we reflect the coordinate system $x \mapsto -x$:

$$-\frac{\mathrm{d}^2x}{\mathrm{d}t^2} - x\frac{-\mathrm{d}x}{\mathrm{d}t} - x = 0 \qquad .$$

Then, when we apply the time reversal $t \mapsto -t$ we find

$$-\frac{\mathrm{d}^2x}{\mathrm{d}t^2} - x\frac{-\mathrm{d}x}{-\mathrm{d}t} - x = -\left(\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + x\frac{\mathrm{d}x}{\mathrm{d}t} + x\right) = 0 \qquad .$$

Therefore, we recover the original expression for the system, implying the time reversibility. Note that this procedure fails when we consider the system representation consisting of coupled first order equations, since the expression for $\dot{y}$ is odd in y.

Figure 4 depicts the phase plane for this dynamical system.

4 Problem 6.7.1: Damped pendulum

Find and classify all the fixed points of

$$\ddot{\theta} + b\dot{\theta} + \sin\theta = 0 \qquad ,$$

for $b > 0$, and plot the phase portraits for the qualitatively different cases.

To do so, we let $\Omega \equiv \dot{\theta}$ and $\boldsymbol{\theta} = \begin{pmatrix}\theta \\ \Omega\end{pmatrix}$. Then, the system can be written in the linearized form

$$\boldsymbol{\theta} \approx \mathbb{A} \cdot \boldsymbol{\theta} \qquad ,$$

where $\mathbb{A}$ is the matrix of stability. The fixed points of the system are given by $(\theta^*, \Omega^*) = (n\pi, 0)$ for $n \in \mathbb{Z}$. For $\dot{\Omega} = -\sin\theta - b\Omega$, nontrivial zeros occur when $\Omega = -\sin\theta/b$. However, since $\dot{\theta} = \Omega$, this is not a fixed point of the system.

To classify the fixed points, we consider the linearized system and the associated matrix of stability (for details see, e.g., section 2.2):

$$\mathbb{A} = \begin{pmatrix} 0 & 1 \\ -\cos\theta & -b \end{pmatrix} \qquad .$$

We consider the case that n is even. This implies that $\cos\theta^* = \cos n\pi = +1$. Hence, $\mathbb{A}$ becomes

$$\mathbb{A} = \begin{pmatrix} 0 & 1 \\ -1 & -b \end{pmatrix} \qquad .$$

Since $\mathrm{Tr}\,\mathbb{A} = -b < 0$, we know immediately that fixed points with this configuration are stable. The discriminant is given by $\mathrm{Tr}\,\mathbb{A}^2 - 4\det\mathbb{A} = b^2 - 4$. Hence, for $0 \leq b < 2$, the behavior around the fixed points are stable spirals. For $b = 2$, the system is degenerate and becomes a star/degenerate node. In the case that $b > 2$, the discriminant is positive and the fixed points become stable nodes.

Now, we consider the case of odd n. Here, $-\cos n\pi = 1$, and we see that

$$\mathrm{Tr}\,\mathbb{A} = -b < 0 \qquad , \qquad \det \mathbb{A} = -1 \qquad .$$

This implies that the fixed points for even n are saddle points, where the eigenvalues of the linearized system near the fixed point are real and have opposite signs.

The qualitatively different phase portraits for the system are depicted for $b = 1, 2, 3$ (from top to bottom) in Figure 5. This shows the transition from stable spirals, to a star node, to a stable node, moving from top to bottom.

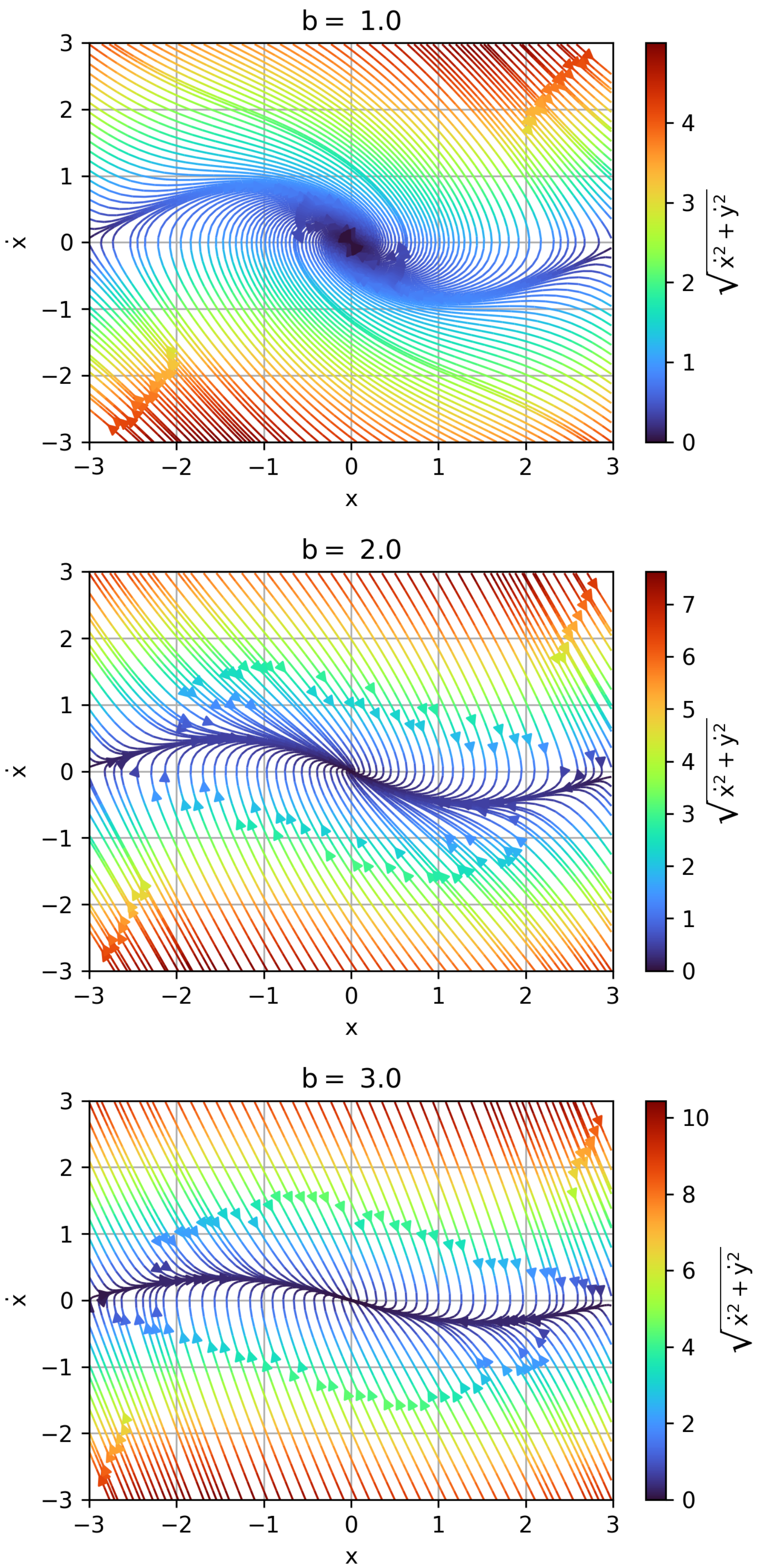

Figure 5: Depiction of various phase portraits for the damped oscillator system in section 4. Panel (a,top) shows the case of $b = 1$, panel (b,middle) shows the critical case for $b = 2$, and panel (c,bottom) displays the system for $b = 3$. These represent the qualitatively unique cases for the phase portrait of this system as a function of b.

5 Software Usage

The entirety of the work was carried out on pen and paper prior to typesetting, except for the plotting: this was completed using python (`numpy` and `matplotlib`).